

Why has digital sovereignty become a strategic question?

Digital Dependence in the Age of AI

Artificial intelligence (AI) is assuming a growing importance in geopolitics. Its role is evolving to the point where it directly impacts how states pursue their domestic and foreign policies. As discussed in the course, AI capability for a state is ultimately a production function of distinct inputs:

$$A = f(C, E, D) \times T \times \eta$$

Here, **C** stands for **compute**, **E** is **energy**, **D** is **data**, **T** is **talent**, and **η** is the **efficiency** with which a given actor can combine them. The states and firms that control these critical inputs of AI are accumulating structural advantages over those that do not, making digital dependence a pressing security concern.

This is not a prospective concern. The **United States** and **China** already dominate most layers of the AI infrastructure stack, from chip design and fabrication to the cloud platforms upon which models are trained and deployed. China controls around 90% of global rare earth refining capacity, essential for the hardware underpinning all digital systems, while Amazon, Microsoft, and Google alone account for roughly 70% of the global cloud market. The models themselves follow the same pattern: 70% of foundational AI systems have been developed in the United States and 15% in China (Bria et al., 2025)¹. For most nations, this means that the chips, data centres, and platforms their economies depend on are designed, owned, and governed elsewhere. This creates deep structural dependencies and clear imbalance in the distribution of AI power.

For those who do not control all these inputs, the question is no longer whether to engage with foreign systems, but how to do so, with whom, on what terms and at what cost. These choices are being made now, shaping innovation, growth, and the modernisation of public services. This strategic context is what leads to the question: ***is it possible for middle power countries to achieve digital sovereignty?***

Defining digital sovereignty

In this analysis, **digital sovereignty** is defined as a state's domestic capacity to exercise meaningful, informed, and politically accountable control over its digital infrastructures, data regimes and governance rules on which collective life is structured upon. For most nations, it involves managing international partnerships to overcome unavoidable external dependencies in line with its own legal frameworks, public interest and strategic priorities.

This definition's framing deliberately focuses on **states**. The infrastructure it describes is largely built and operated by private actors whose decisions carry geopolitical consequences. That dimension is not ignored here but sovereignty, as a legal and political concept, remains a state function.

¹ https://assets.filedock.xyz/public/EuroStack_2025.pdf

Digital Sovereignty as Defined by International Organizations

No major intergovernmental body has adopted a single binding definition of digital sovereignty, and none explicitly uses the term as a formal concept. The institutions chosen represent broad and authoritative cross-sections of intergovernmental thinking on digital governance. As such, they offer a set of adjacent frameworks that can help address the problem from different entry points.

The **OECD** approaches the issue from the angle of state capacity. In its **Digital Government Policy Framework (2020)**², it argues that governments need more than online services: they need public institutions that can actually design, manage, and adapt digital systems effectively. In this view, digital sovereignty begins with the ability of the state itself to govern in a digital environment, rather than simply putting existing bureaucratic procedures online.

UNCTAD, through its **Digital Economy Report: Shaping an Environmentally Sustainable and Inclusive Digital Future (2024)**, approaches the issue from a different angle³. It stresses that digital infrastructure depends on scarce material and energy inputs and that developing countries risk remaining marginalized in digital value chains unless governance and investment patterns change. In this framing, the key issue is not sovereignty itself but data governance and the structural terms under which countries participate in the digital economy.

The **UN's Global Digital Compact (GDC)** (2024) sets out a framework for an open, free and secure digital future for all, grounded in international cooperation, inclusion, rights and governance⁴. It emphasizes building global conditions for digital technologies, focusing on multi-stakeholder processes, connectivity, human rights and equitable access. Notably, the GDC recognizes that multiple implementation models can coexist and must be adapted to different contexts, according to local needs, capacities and priorities.

The **World Bank** approaches the question through its work on **Digital Public Infrastructure and Development Report (2025)**, defining DPI as shared, open digital foundations designed to reduce dependence on proprietary systems while enabling inclusive public service delivery⁵. Rather than articulating a doctrine of digital sovereignty, it reflects a practical concern with maintaining public control over essential digital systems.

Takeaway: These frameworks do not provide a single agreed definition of digital sovereignty, but they do point in the same direction. *All converge on the same underlying concern: ensuring that states retain enough agency to make meaningful choices about the digital systems that structure public and economic life, rather than simply inheriting arrangements designed by and for others.*

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https://www.oecd.org/en/publications/the-oecd-digital-government-policy-framework_f64fed2a-en.html

³ https://unctad.org/system/files/official-document/der2024_en.pdf

⁴ <https://unglobalcompact.org/library/6240>

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<https://documents1.worldbank.org/curated/en/099031025172027713/pdf/P505739-84c5073b-9d40-4b83-a211-98b2263e87dd.pdf>

Debates Around the Concept of Digital Sovereignty

Because the term lacks a universally accepted definition, it is contested and justified in a myriad of ways across policy debates globally. The term “digital sovereignty” is often invoked by actors pursuing fundamentally incompatible projects, such as states asserting infrastructure control, firms marketing compliance products, and movements resisting corporate surveillance, with no shared metric to determine when any of them has succeeded⁶. Pohle and Thiel (2020) observe that the sovereignty discourse tends to appear most forcefully precisely where authority is weakest, making it as much a political speech act as a policy programme⁷. Santaniello (2025) argues that the resulting gap between rhetoric and implementation is an inherent attribute of sovereignty claims⁸. States can keep invoking digital sovereignty even when their policies fail to deliver it because the claim itself serves political functions by mobilising support, signalling intent, and framing national identity, which are independent of whether the underlying condition is actually achieved. The concept is politically self-sustaining even in the face of its own shortcomings.

Beyond its kaleidoscopic definitions, full digital sovereignty in practice – understood as unilateral control over the entire digital stack – is a condition currently fulfilled by no single state. The concept thus largely remains an unattainable objective. In such a reality, the key question is not whether a state can retain full control. The sophistication of the question lies in **whether a state can still make meaningful choices about where data flows, whose infrastructures shape public life, what legal protections apply and how dependencies are managed**. Digital sovereignty is less about total independence than about governance under conditions of constraint. It is precisely because these inputs are finite, unevenly distributed, and increasingly contested that digital sovereignty has become an operational priority for states. Because states must operationally position themselves and define their AI strategies in response to concrete physical constraints, digital sovereignty retains meaning regardless of the theoretical ambiguity that surrounds it.

The Concept of Middle Power and Its Strategic Dilemmas

As Canadian Prime Minister Mark Carney remarked at the World Economic Forum in Davos in 2026: *“When we [middle powers] only negotiate bilaterally with a hegemon, we negotiate from [a position of] weakness... This is not sovereignty. It’s the performance of sovereignty while accepting subordination... In a world of great power rivalry, the countries in between have a choice: compete with each other for favour, or combine to create a third path with impact. We shouldn’t allow the rise of hard power to blind us to the fact that the power of legitimacy, integrity and rules will remain strong if we choose to wield it together.”*⁹

While great powers can absorb the costs of technological rivalry more easily because they control key firms, infrastructures, and strategic chokepoints, smaller states often have little choice but to adapt to their external circumstances. Middle powers sit in between these two poles. They are integrated enough in the global digital economy to be a commercially

⁶ <https://doi.org/10.1177/1461444819865984>

⁷ <https://doi.org/10.14763/2020.4.1532>

⁸ <https://doi.org/10.1080/14650045.2025.2521548>

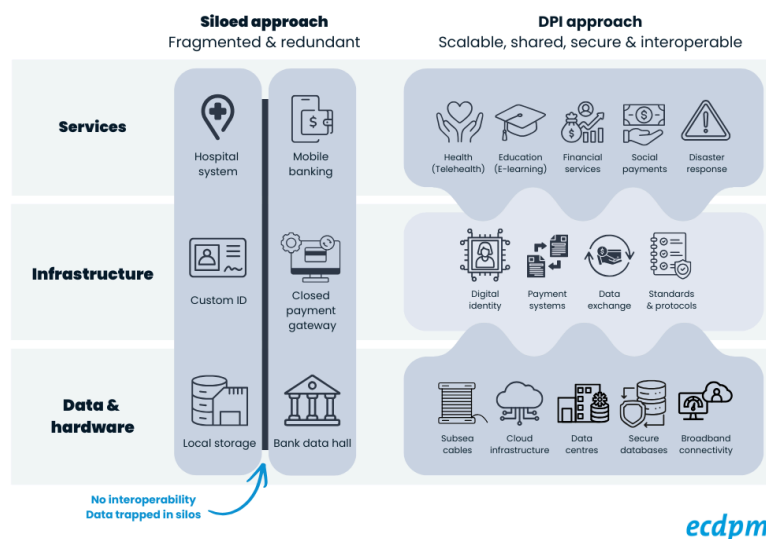
⁹ <https://www.weforum.org/stories/2026/01/davos-2026-special-address-by-mark-carney-prime-minister-of-canada/>

meaningful market that deploys this infrastructure, yet they lack the resources to fully develop them at home.

Analysis of Assets and Vulnerabilities: Shaping Middle Power Approaches to Digital Sovereignty

States that fall under this definition of a middle power are typically shaped by varying levers, resources, and constraints, despite navigating similar strategic choices. With enough room to act strategically but not enough to combat all dependencies, their strategies are a result of their specific combination of assets and vulnerabilities.

Those assets can be organised into five broad categories. The first is **market power**, which considers the size and affluence of the domestic market. This examines to what extent a state can use access to its market as a compliance lever over foreign firms. The second is **regulatory and normative authority**. This category focuses on state capacity to set rules that others follow as a condition of market access, which can substitute for hardware control when enforced at sufficient scale.



Source: ECDPM, From Brazil to the World: Open Digital Infrastructure for Climate Cooperation, 2026

The third is **digital public infrastructure (DPI)**. States that have built open, interoperable DPI at scale create public digital rails (normally consisting of identity, payment and data exchange systems) on which both public and private applications can run. This reduces dependence on closed proprietary systems and limits the lock-in that accompanies such structures.

The fourth is **physical endowments**, which considers energy availability, renewable capacity, critical mineral reserves, and geographic positioning. Such physical endowments determine which parts of the AI stack a state can credibly claim to support or supply. The fifth considers **diplomatic and institutional positioning**, which focuses on the state's ability to use multilateral forums, standard-setting bodies, and bilateral partnerships to shape governance norms and diversify dependencies, even in the absence of independent productive capacity.

Their dilemma, then, is how to manage strategic dependencies. They must decide which partnerships to pursue, which vulnerabilities to reduce and where to build leverage. The strategies they adopt are therefore analytically understood as a result of the specific combination of assets and vulnerabilities of each middle power. This can take different forms, as examined through the case studies of **Brazil**, the **European Union**, **India** and **Kenya**.

CASE STUDIES

The following case studies examine how middle powers operationalise digital sovereignty under conditions of structural dependence. Rather than treating digital sovereignty as a single policy model to be replicated, the cases are analysed as different configurations of the same underlying problem: *how states that lack full control over the AI stack manage dependence, build leverage where possible, and prioritise specific layers of the digital ecosystem according to their economic, developmental, and geopolitical objectives.*

Each case mobilises a distinct combination of assets while accepting persistent vulnerabilities. Read comparatively, the cases illustrate that digital sovereignty for middle powers is not a question of absolute autonomy, but of strategic choice: deciding where sovereignty matters most, which dependencies are tolerable, and how leverage can be accumulated over time within the constraints of an unequal global digital order.

The European Union — Regulating Power in a World It Does Not Build

The European Union (EU)'s strategy has long relied on its ability to leverage its large, affluent single market to compel foreign firms to comply with European regulatory standards, a mechanism known as the Brussels Effect¹⁰. As a result, instruments such as the General Data Protection Regulation (GDPR), the Digital Markets Act (DMA), and the Digital Services Act (DSA) extend beyond consumer protection alone to function as tools of normative influence, turning the EU into the world's foremost regulatory power in the digital domain.

However, this influence rests on foundations that risk weakening. The EU's share of global GDP has fallen from 30 percent in 2008 to 21 percent today and is projected to reach 16 percent by 2050, directly eroding the cost-of-bypass calculation that historically made compliance with EU standards the rational choice for global firms. While GDPR became a global privacy standard through the Brussels Effect, the AI Act has not yet achieved equivalent reach¹¹. Its enforcement is inconsistent across member states and jurisdictions that adopted GDPR-aligned frameworks are now taking a wait-and-see approach before following Europe's lead again^{12 13}.

Overall, the EU's approach reveals an appetite for governing technologies it does not fully control – a paradox sitting at the heart of the European digital project. Now, the bloc is trying to move beyond regulation alone, using industrial policy and external partnerships to reduce critical dependencies and strengthen its position across other layers of the digital stack¹⁴¹⁵. This tension makes the EU a particularly instructive case for understanding digital sovereignty not as a fixed objective to be attained, but as a dilemma to be continuously managed.

¹⁰ <https://global.oup.com/academic/product/the-brussels-effect-9780190088583>

¹¹ <https://global.oup.com/academic/product/the-brussels-effect-9780190088583>

¹² <https://artificialintelligenceact.eu/national-implementation-plans/>

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¹⁴ https://commission.europa.eu/topics/competitiveness/ai-continent_en

¹⁵ <https://digital-strategy.ec.europa.eu/en/policies/international-digital-strategy>

Leverage Points

- **Regulatory market power and standard-setting:** A market of 450 million people and around €18 trillion in GDP is too large to ignore¹⁶. Thus, any firm interested in access to the EU market must meet certain compliance obligations. The GDPR became a global privacy standard; the AI Act is now shaping AI governance conversations far beyond Europe¹⁷. This underscores substantial rule-shaping authority over technologies the EU does not produce. Moreover, by embedding values (privacy, transparency, interoperability) into legally binding frameworks, the EU shapes what "legitimate" AI looks like internationally.
- **The semiconductor chokepoint:** The Dutch firm ASML holds a near-monopoly on extreme ultraviolet (EUV) lithography machines, tools needed to manufacture the world's most advanced chips. No country, including the US or China, can produce cutting-edge semiconductors without them. This is Europe's single most powerful hardware chokepoint in the global AI supply chain.
- **Energy assets:** Although the EU is far from energy independent, it does have meaningful sources of low-carbon electricity, the importance of which increases as geopolitical volatility continues to put pressure on oil and gas supplies. France's nuclear fleet provides large volumes of firm-power electricity attractive to data-intensive AI infrastructure, with the country exporting a record 90 TWh of decarbonised electricity in 2024¹⁸ ¹⁹. These assets are increasingly strategic as global data centre electricity demand surged 17 percent in 2025 and is projected to grow from 460 TWh in 2024 to over 1,000 TWh by 2030²⁰ ²¹. However, these assets are national and not collectively pooled, with no common procurement mechanism to convert them into shared digital infrastructure.
- **Market size and capital mobilisation:** The AI Continent Action Plan (2025) commits €200 billion for AI development, including €20 billion via the InvestAI facility for up to 5 gigafactories and 19 AI factories already operational, targeting a tripling of EU data centre capacity within 5 to 7 years ²² ²³ ²⁴. The financing architecture has been confirmed by the European Investment Bank²⁵.
- **Frontier Model Capability - Proof of Concept:** Mistral AI, founded in April 2023 by three French researchers, has grown to a valuation exceeding \$14 billion and raised \$830 million in March 2026 to build a 44 MW data centre near Paris, with a total

¹⁶ https://european-union.europa.eu/principles-countries-history/facts-and-figures-european-union_en

¹⁷ <https://www.sciencedirect.com/science/article/pii/S2666659625000241>

¹⁸ <https://www.datacenterdynamics.com/en/analysis/france-ai-data-center-build-out-emmanuel-macron/>

¹⁹ <https://www.iea.org/news/data-centre-electricity-use-surged-in-2025-even-with-tightening-bottlenecks-driving-a-scramble-for-solutions>

²⁰ <https://www.iea.org/news/data-centre-electricity-use-surged-in-2025-even-with-tightening-bottlenecks-driving-a-scramble-for-solutions>

²¹ <https://www.iea.org/reports/energy-and-ai/energy-supply-for-ai>

²² https://commission.europa.eu/topics/competitiveness/ai-continent_en

²³ <https://digital-strategy.ec.europa.eu/en/policies/ai-factories>

²⁴ <https://digital-strategy.ec.europa.eu/en/factpages/ai-continent-action-plan>

²⁵ <https://www.eib.org/en/press/all/2025-491-eib-group-and-european-commission-join-forces-to-finance-ai-gigafactories>

European infrastructure budget of €4 billion²⁶. It functions as Europe's main operational frontier AI company with dedicated sovereign compute infrastructure under construction²⁷. It holds important European significance, but its primary identity and regulatory home is French.

- **Human capital:** Europe boasts strong universities with 37.22% of the population having completed tertiary education and certain countries having even higher rates than the United States²⁸. However, students must be mobilised accordingly, as presently only a little over a quarter pursue degrees in STEM fields. European talent can produce frontier-class models, as evidenced through Mistral AI, but it still needs to be incentivised and retained²⁹.
- **Diversification through partnerships:** The EU's 2025 International Digital Strategy is Europe's attempt to expand beyond the Brussels Effect³⁰. It seeks to combine regulation, public-private investment and digital cooperation through a "Team Europe" approach³¹. Global Gateway is a key instrument of this shift, using infrastructure finance and digital partnerships to increase connectivity, cloud, data and standards cooperation abroad. This helps the EU diversify partners, secure digital governance around EU values and project influence beyond its own market. Furthermore, the EU's ongoing digital trade agreement negotiations with Canada, mutual data adequacy agreement with Brazil, and deepening digital economy collaboration with India reflects the EU's approach to seek like-minded partnerships that align with their terms^{32 33 34}.

Main Dependencies / Limits

- **Market domination by foreign cloud providers:** European cloud providers hold less than 15% of their domestic market, with that share shrinking over time³⁵. Some of the largest players in this space, AWS, Azure and Google Cloud, are US companies subject to the US CLOUD Act³⁶. This is particularly poignant as the Act compels US-companies to comply with warrants or subpoenas, regardless of the geographical location of the data stored.
- **Fragmented internal market for digital:** With 24 official languages and 27 member states with divergent industrial strategies, energy policies and risk tolerances, collective action is slow and costly.
- **Shallow capital markets:** European capital markets pale in comparison to those in the United States, with US-listed companies achieving valuations 3.32 times higher

²⁶ <https://mistral.ai/about>

²⁷ <https://europe.mistral.ai/>

²⁸ <https://www.oecd.org/en/data/indicators/population-with-tertiary-education.html>

²⁹ <https://education.ec.europa.eu/resources-and-tools/data-and-analysis-on-education-and-skills/key-trends-in-education/stem-in-tertiary-education#:~:text=Across%20the%20EU%2C%20just%20over,target%20of%20at%20least%2032%25>

³⁰ https://ec.europa.eu/commission/presscorner/detail/en/ip_25_1370

³¹ https://international-partnerships.ec.europa.eu/policies/team-europe-initiatives_en

³² https://ec.europa.eu/commission/presscorner/detail/en/ip_26_553

³³ https://ec.europa.eu/commission/presscorner/detail/en/ip_26_229

³⁴ https://policy.trade.ec.europa.eu/eu-trade-relationships-country-and-region/countries-and-regions/india/eu-india-agreements/memo-eu-india-free-trade-agreement-chapter-chapter-summary_en

³⁵ <https://www.cnbc.com/2026/02/13/four-charts-europes-reliance-us-digital-infrastructure.html>

³⁶ <https://www.alstonprivacy.com/cloud-act-impact-cross-border-access-contents-communications/>

on average than their European counterparts³⁷. Moreover, Europe's venture capital (VC) market lags behind that of the United States³⁸. Between 2014 and 2023, VC investment amounted to EUR 89 billion in the EU, compared with over EUR 1,000 billion in the United States. This lower-liquidity environment disincentivises domestic entrepreneurship, particularly in a sector as capital intensive as AI. The Draghi Report identified the EU venture ecosystem's fragmented, risk-averse nature as a structural weakness³⁹.

- **ASML's chokepoint is not permanent:** China is investing heavily in domestic alternatives to EUV lithography. The lead Europe holds through ASML is real but time-limited.
- **Energy Dependency:** Despite progress in green generation, the continent is not self-sufficient overall. In 2024, it still imported 57% of its energy needs⁴⁰, with major dependencies on oil and gas suppliers. This presents an indirect but real vulnerability for further infrastructure build out, as electricity systems fall under increasing strain due to growing power demand and bottlenecks, either caused internally or externally.

Overall, the EU's strategy is a calculated attempt to convert regulatory leverage into productive capacity before the former erodes. Brussels is simultaneously trying to do multiple things at once: retain its normative authority through regulation and partnerships while preventing regulatory complexity from becoming a competitive liability and innovation barrier through simplification initiatives and asset leveraging. As such, the EU is using its legal weight to shape the rules of the game while simultaneously trying to diversify and build enough productive capacity to play. Whether it can do both simultaneously without losing rule-making credibility in the process is the central strategic question.

Why this strategy? Usage of Assets:

Since the EU lacks domestically dominant hyperscalers and frontier AI firms, its most credible source of influence lies not in production but in rule-setting. This structural constraint explains why the EU prioritises **regulatory and normative authority** and **market power**, while starting to expand its **diplomatic and institutional positioning**, as substitutes for hardware and platform control, rather than attempting immediate technological parity with the United States or China.

Kenya — The Sovereignty Paradox of the Silicon Savannah

Kenya is a small, lower-middle-income East African economy of 55 million people that has become the most advanced digital economy in East Africa and one of the continent's clearest cases of digital sovereignty pursued through visibility rather than ownership. **Kenya's National AI Strategy 2025–2030** calls for AI development centres, a national

³⁷ <https://www.ecb.europa.eu/pub/pdf/scpops/ecb.op373.en.pdf>

³⁸ <https://www.banque-france.fr/en/publications-and-statistics/publications/how-can-europe-scale-its-venture-capital-market#:~:text=However%2C%20Europe's%20VC%20market%20still,stages%20of%20start%20Dup%20financing.>

³⁹ https://commission.europa.eu/document/download/97e481fd-2dc3-412d-be4c-f152a8232961_en

⁴⁰ <https://ec.europa.eu/eurostat/web/interactive-publications/energy-2026>

dataset initiative, localised AI models, and governance reform⁴¹. Kenya is effectively converting the credibility of its DPI ecosystem into geopolitical positioning, while remaining structurally reliant on the foreign capital and foreign platforms that make that ecosystem run.

Leverage Points

- **M-Pesa and DPI as geopolitical currency:** M-Pesa reaches over 90% of Kenyan adults and processes over 70% of the country's digital payment transactions. The government has also developed eCitizen, a portal hosting 16,000 public services, and NOFBI, a national fibre backbone connecting all 47 counties^{42 43}.
- **Renewable energy comparative advantage:** Kenya has a relatively strong electricity base for digital infrastructure. In 2023/24, 83% of the electricity supplied to the national grid came from renewable sources, and geothermal alone met 41.71% of total electricity demand, providing stable baseload power that many African grids cannot match. This helps explain why Microsoft and Abu Dhabi-based technology firm G42 announced a new East Africa cloud region in Kenya, with the data centre expected to be powered by geothermal energy. It is one of Kenya's clearest structural advantages for data centre development⁴⁴.
- **PAPSS integration and regional financial positioning:** AI for development is central in Kenya, where digital systems are closely tied to public service delivery and economic transformation. The February 2026 integration of Kenya's PesaLink into the Pan-African Payment and Settlement System (PAPSS) connects over 80 Kenyan financial institutions to 160-plus institutions across Africa in real-time local-currency settlement, bypassing dollar-dominated correspondent banking⁴⁵. Afreximbank estimates \$5 billion in continental savings from full PAPSS rollout.
- **ICT sector momentum attracting regional human capital:** Kenya's ICT sector has grown at 10.8 percent annually over the past decade⁴⁶. Kenyan startups attracted close to \$800 million in funding in 2023. The country attracts intra-African tech talent, including engineers from Nigeria and Uganda.
- **Diplomatic positioning and AU institutional role:** Kenya has successfully claimed the role of Africa's "digital capital" by hosting PAPSS headquarters, occupying the AU Champion on Institutional Reform role, and parlaying its DPI track record into a seat at global AI governance conversations despite its comparably lower productive capacity.

Main Dependencies / Limits

- **Compute is foreign-owned:** The Olkaria campus, announced in May 2024 as Kenya's flagship AI infrastructure project and to be built by Microsoft and UAE-based

⁴¹ <https://ict.go.ke/sites/default/files/2025-03/Kenya%20AI%20Strategy%202025%20-%202030.pdf>

⁴² <https://www.m-pesa.africa/>

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https://accounts.ecitizen.go.ke/en?__cf_chl_rt_tk=_vaNVrukXyftSd4KcTND2IRxFDBPOwJxbsZiPBVb_bBM-1778138672-1.0.1.1-ZoTo6eYZ_N6XKMHQhkgmslJ9vJvHJ3UvBGs8YMvg6Os

⁴⁴ https://www.epra.go.ke/sites/default/files/2024-10/EPRA%2520Energy%2520and%2520Petroleum%2520Statistics%2520Report%2520FY%25202023-2024_2.pdf?&sa=D&source=docs&ust=1778142306943049&usg=AOvVaw1hGTzyRtlxASflJ0Gogke_

⁴⁵ <https://pesalink.co.ke/>

⁴⁶ <https://www.trade.gov/country-commercial-guides/kenya-digital-economy>

G42 on KenGen's geothermal land in Naivasha, has stalled due to grid capacity constraints and the absence of National Treasury financial approval. Kenya's only domestic semiconductor initiative, the STL nanofabrication facility at Dedan Kimathi University in Nyeri, produces legacy chips for testing and research rather than the advanced processors that AI infrastructure requires. A \$1.3 million US feasibility grant awarded to STL in May 2024 under the Biden administration's CHIPS Act to study a potential fabrication facility has an uncertain future, as the Trump administration has redirected international semiconductor funding toward domestic US production⁴⁷⁴⁸⁴⁹⁵⁰. Kenya currently relies on imports of advanced semiconductors to power its AI ambitions. Supporters of Kenya's semiconductor ambitions may lean on the country's mineral wealth as a pillar for semiconductor manufacturing. However, most of the value in the semiconductor industry lies far beyond extraction, in processing and fabrication.

- **State data sits on foreign clouds:** Despite the Cloud-First Policy's localist framing, sensitive government databases are nonetheless stored on Microsoft Azure, AWS, and Huawei Cloud⁵¹. The National Cloud Data Centre at Konza Technopolis, whose purpose was to address this risk, remains underutilised, symbolising the gap between policy aspiration and operational reality.
- **Weak data governance:** The Data Protection Act of 2019 established the Office of the Data Protection Commissioner as the primary oversight body, but its office is under-resourced and administrative fines are capped at approximately \$50,000, low enough that compliance costs often exceed penalty risk for large platforms. The government has simultaneously expanded surveillance powers via the October 2025 Computer Misuse and Cybercrimes Amendment Act, creating a legitimacy gap between sovereignty claimed externally and rights protected internally⁵²⁵³.
- **No binding AI legal framework:** As of early 2026, no AI-specific legislation is in force. The Robotics and AI Society Bill remains pending. The National AI Strategy functions as a political document, not an enforceable framework⁵⁴⁵⁵.
- **Talent drain with no retention mechanism:** Kenya's tech ecosystem and position as the Silicon Savannah produces strong engineers and data scientists, but lacks a binding retention mechanism, diaspora equity programme, and or return-migration incentive. Silicon Valley, London, and Gulf tech hubs consistently absorb Kenyan talent at compensation levels the domestic market cannot match.

⁴⁷<https://www.semafor.com/article/05/06/2026/energy-shortfall-problem-scuppers-kenyas-1b-microsoft-data-center>

⁴⁸<https://www.dkut.ac.ke/index.php/launch-of-the-machine-building-division-of-semiconductor-technologies-limited>

⁴⁹<https://www.ustda.gov/ustda-partners-with-kenya-to-boost-semiconductor-manufacturing/>

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<https://www.manufacturingdive.com/news/trump-biden-chips-act-future-federal-cuts-layoffs-musk/741052/>

⁵¹<https://www.ict.go.ke/sites/default/files/2024-12/Kenya%20Cloud%20Policy%20-%202024.pdf>

⁵²<https://www.kentrade.go.ke/wp-content/uploads/2022/09/Data-Protection-Act-1.pdf&sa=D&source=docs&ust=1778142674933664&usg=AOvVaw3lBklsJBAflWWn2gswl8Er>

⁵³<https://www.parliament.go.ke/sites/default/files/2025-10/Computer%20Misuse%20and%20Cybercrimes%20%28Amendment%29%20Act%2C%202025.pdf>

⁵⁴https://www.dataguidance.com/sites/default/files/the_kenya_robotics_and_artificial_intelligence_society_bill_2023.docx.pdf

⁵⁵<https://ict.go.ke/sites/default/files/2025-03/Kenya%20AI%20Strategy%202025%20-%202030.pdf>

Kenya's actual strategy is **DPI-as-diplomatic-leverage**: use the credibility of operational public digital infrastructure, built on real, functioning systems like M-Pesa, PesaLink and eCitizen to claim governance authority and attract foreign investment, while managing rather than resolving the underlying structural dependencies. Kenya is not building sovereign AI; it is building the institutional profile of a digital leader, which then gives it access to partnerships, headquarters locations, and governance forums that generate secondary advantages. The gap between hosting infrastructure and owning it remains unresolved. Thus, Kenya's case helps demonstrate that visibility and ownership are two very different things.

Why this strategy? Usage of Assets:

For Kenya, digital sovereignty is shaped by the constraints of a small market and limited capital, which make ownership of frontier AI infrastructure unrealistic in the short term. As a result, the state prioritises visibility, credibility, **physical endowments** and **diplomatic positioning** through **operational DPI**, using these assets to attract investment and governance influence rather than attempting costly technological self-sufficiency.

India — The Greenfield Sovereign That Still Imports Its Stack

India frames digital sovereignty through an explicitly techno-nationalist lens. The "Atmanirbhar Bharat" (self-reliant India) doctrine links technological independence to national security, economic growth, and an anti-colonial narrative drawing explicit parallels between historical data extraction by US tech firms and the resource extraction of British colonial rule⁵⁶. Thus, India's approach to digital sovereignty, among other things, includes unrolling its **India Stack**: a layered system of DPI built from scratch in the 2010s. India's DPI has delivered measurable results in a short time span by raising the country to the top of the digitalisation index of revenue administrations among emerging market economies, while transforming social service delivery across education, health, and agriculture through platforms like DIKSHA, ABDM and VISTAAR^{57 58 59}. This "**greenfield**" model of designing systems digitally from the outset rather than patching legacy ones has been India's most reproducible export.

Leverage Points

- **Operational Homegrown DPI**: India has built Aadhaar (biometric identity for over 1.3 billion people), UPI (now the world's fifth-largest payment network by volume, accounting for 70 percent of India's digital payment transactions), and DigiLocker (digital document storage and verification)^{60 61 62}. UPI is live in the UAE, Bhutan, Oman, and Singapore; pilot-tested in France, Nepal, Mauritius, and Sri Lanka. The

⁵⁶ <https://tribal.nic.in/atmanirbhar-bharat.aspx>

⁵⁷ <https://www.indiastack.global/diksha/>

⁵⁸ <https://www.india.gov.in/spotlight/ayushman-bharat-digital-mission-abdm>

⁵⁹ <https://vistaar.da.gov.in/>

⁶⁰ <https://uidai.gov.in/en/>

⁶¹ <https://www.npci.org.in/product/upi>

⁶² <https://www.digilocker.gov.in/>

Modular Open Source Identity Platform (MOSIP), a non-profit initiative, hosted by the International Institute of Information Technology, Bangalore (IITB), is already adopted by over twenty countries and has more than 121 million active users⁶³.

- **Scale of domestic market:** With 1.4 billion people and rapidly expanding digital access, India offers a market large enough to sustain domestically-oriented platforms and justify investments that smaller economies could not. According to the Nasscom-Arthur D Little “**Digital Public Infrastructure of India - Accelerating India’s Digital Inclusion**” report, the economic value added from DPIs to India’s GDP is estimated to reach 2.9 - 4.2% by 2030 (from 0.9% in 2022)^{64 65}.
- **Public compute through the IndiaAI Mission:** The IndiaAI Mission, launched in 2023, is a government initiative to build a comprehensive ecosystem to foster AI innovation across various sectors in India⁶⁶. Spearheaded by the Ministry of Electronics and Information Technology (MeitY), it focuses on developing AI applications to address societal challenges in healthcare, education, agriculture, and smart cities while promoting responsible and ethical AI development. It is responsible for deploying more than 34,000 Graphics Processing Units (GPUs) by 2025, creating one of the largest public AI compute networks globally. This reduces Indian start-ups’ and researchers’ dependence on private US hyperscalers, creating a partial buffer between domestic AI development and foreign compute dependence.
- **Bhashini and multilingual data sovereignty:** The Bhashini initiative develops natural language processing models for India’s 22 official languages using open-source frameworks, with datasets made freely available to developers⁶⁷. This addresses a structural gap in global AI. Most frontier models perform poorly on non-English languages while building an Indian-language AI asset that foreign models cannot easily replicate.
- **Talent depth and global positioning:** According to the 2025 Global Workplace Skills Study by Emeritus, 96% of Indian professionals use AI tools at work, significantly above the 81% US rate⁶⁸. India’s IITs and IISc actively contribute to global open-source AI research. India is a major destination for global companies seeking skilled AI professionals, giving it bargaining power in global tech investment negotiations.
- **Strategic positioning for semiconductors and nuclear energy:** India is developing its own semiconductor manufacturing capacity under the India Semiconductor Mission (2021), and its long-term carbon-neutrality roadmap (net-zero by 2070) is driving investment in nuclear baseload power (via the SHANTI Bill) specifically designed to support AI infrastructure by supporting India’s long-term

⁶³ <https://www.mosip.io/>

⁶⁴ <https://nasscom.in/knowledge-center/publications/nasscom-arthur-d-little-indias-digital-public-infrastructure>

⁶⁵ <https://www.institutmontaigne.org/en/expressions/indias-digital-public-infrastructure-success-story-world>

⁶⁶ <https://indiaai.gov.in/>

⁶⁷ <https://bhashini.gov.in/>

⁶⁸ <https://emeritus.org/in/global-workplace-skill-study-2025-india-edition/>

vision of **expanding clean, reliable energy**^{69 70}. India has set an ambitious target of achieving 100 GW of nuclear capacity by 2047.

- **Building diplomatic and supply-chain partnerships:** India signed MoUs with Australia, Argentina, Brazil and the DRC to secure lithium and cobalt^{71 72 73 74}. It joined the G7-led Mineral Security Partnership (MSP) in 2023, tapping into global networks for funding and recycling, and with the EU it has a Trade and Technology Council^{75 76}.
- **DPI as global norm export:** India used its G20 presidency in 2023 to globalise the DPI concept by launching the Global Digital Public Infrastructure Repository (GDPIR), a virtual platform for exchanging DPI knowledge and best practices at scale, to which Australia, France, Singapore, and Oman have voluntarily contributed. Beyond this, India has signed bilateral MoUs with eleven countries across the continents including Kenya, Colombia, and Tanzania to support their digital transformation.

Main Dependencies / Limits

- **Semiconductor and critical minerals import dependence:** India imports over 95 percent of its semiconductors, meaning its entire AI ambition rests on a supply chain it does not control⁷⁷. The consequences are real: when Microsoft temporarily blocked services to Nayara Energy in 2025, a single corporate decision made abroad disrupted operations at a major Indian company with no domestic recourse. India remains largely dependent on imports for energy transition minerals and their compounds, with 100% import dependency for minerals like lithium, cobalt and nickel, even though it has sizable reserves of cobalt (44.9 Mt ore), nickel (189 Mt), copper (163.9 Mt), and graphite^{78 79}.
- **Foreign cloud infrastructure:** Microsoft, Amazon, and Google collectively announced over \$67 billion in India investments but they bring capacity, not sovereignty. Data centres built by foreign hyperscalers in Hyderabad, Mumbai, and Pune remain subject to US jurisdiction, including the CLOUD Act.

⁶⁹ <https://ism.gov.in/>

⁷⁰ <https://static.pib.gov.in/WriteReadData/specificdocs/documents/2025/dec/doc20251222741701.pdf>

⁷¹

<https://ieefa.org/resources/critical-minerals-india-must-step-its-strategies#:~:text=Recognising%20the%20urgency%20of%20securing,investment%20in%20mineral%20supply%20chains>

⁷² https://www.mea.gov.in/Uploads/PublicationDocs/33569_MEA_annual_Report.pdf

⁷³ <https://www.mea.gov.in/Portal/LegalTreatiesDoc/BR25B4580.pdf>

⁷⁴ <https://ieefa.org/resources/critical-minerals-india-must-step-its-strategies#:~:text=Recognising%20the%20urgency%20of%20securing,investment%20in%20mineral%20supply%20chains>

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<https://2021-2025.state.gov/minerals-security-partnership/#:~:text=The%20MSP%20aims%20to%20ensure,countries%20developing%20their%20minerals%20sectors>.

⁷⁶

<https://digital-strategy.ec.europa.eu/en/news/eu-and-india-deepen-strategic-engagement-second-trade-and-technology-council>

⁷⁷

<https://www.bluekraft.in/wp-content/uploads/2024/12/The-Beginning-of-Indias-Semiconductor-Age-01.pdf>

⁷⁸ <https://ieefa.org/sites/default/files/2024-10/India%27s%20Hunt%20for%20Critical%20Minerals.pdf>

⁷⁹

<https://ieefa.org/resources/critical-minerals-india-must-step-its-strategies#:~:text=dependent%20on%200imports%20for%20energy,supply%20of%20critical%20minerals%20through>

- **Talent drain to Silicon Valley:** India produces exceptional AI engineers and exports a disproportionate share of them. The Atmanirbhar rhetoric of self-reliance has not been matched by policies capable of retaining the talent needed to make it real. The IIT-to-Silicon Valley pipeline remains one of the most persistent structural flows in global tech.
- **Data protection lagging behind data localisation:** India has pushed for data localisation through the Digital Personal Data Protection (DPDP) Act, but its data protection standards remain significantly weaker than the EU's GDPR or China's Data Security Law⁸⁰. The gap between asserting data sovereignty and actually protecting the data it governs is a credibility problem for India's model.
- **The open-source dilemma.** The practical path to AI capability for India runs through fine-tuning Meta's open-source Llama models rather than building foundation models from scratch⁸¹. This is faster and cheaper but it means India's AI "sovereignty" rests on a foundation it did not build and does not control, undermining the Atmanirbhar narrative yet again.

India's strategy is best described as **greenfield DPI with selective sovereignty**: build from scratch where it is strategically viable (identity, payments, language infrastructure) and accept managed dependence where it is not (chips, cloud, and foundation models). But the contradictions between Atmanirbhar rhetoric and actual import dependency are persistent and visible. India uses the language of sovereignty to mobilise political support and attract investment. The goal is not to escape dependence but to accumulate enough domestic capability to negotiate its terms from a position of strength.

Why this strategy? Usage of Assets:

India's prioritisation of **digital public infrastructure** is driven less by industrial competition with frontier AI powers than by the need to scale public service delivery across a vast population with uneven state capacity. In this context, sovereignty at the application and governance layers matters more politically than sovereignty over chips or foundation models. This explains India's willingness to accept managed dependence at the hardware layer, leveraging its **market power** to attract foreign investment and using selective **diplomacy positioning** to secure inputs, diversify dependencies, and extend the reach of its DPI model.

Brazil — The Interdependent Brownfield State: Reusing Systems, Diversifying Partners

Rather than asserting independence from foreign infrastructure or building a new digital ecosystem from scratch, Brazil has pursued what its government explicitly calls a "**brownfield**" strategy: upgrading, connecting, and repurposing existing legacy systems into a unified digital public infrastructure, while simultaneously using multilateral diplomacy to diversify external dependencies. This is tied to a broader agenda of state transformation, in which digital infrastructure is linked to social, economic and environmental missions defined through interministerial plans, with the **Ministry of Management and Innovation in Public**

⁸⁰ <https://static.pib.gov.in/WriteReadData/specificdocs/documents/2025/nov/doc20251117695301.pdf>

⁸¹ <https://www.llama.com/>

Services (MGI) playing a coordinating role. The National Strategy for Digital Government 2024–2027 formalises this direction by emphasising inclusion, interoperability, secure data reuse, and simpler access to services across federal, state, and municipal governments⁸².

Leverage Points

- **Operational DPI at scale:** Gov.br (160 million users, 12,000+ integrated public services), Pix (150 million+ users, Brazil's dominant payment method), and Conecta Gov.br (securing information-sharing across public systems) form the most operationally complete DPI ecosystem outside of India and China^{83 84 85}. These are running, widely adopted, and generating reusable public datasets.
- **Climate data as a sovereign asset:** Cadastro Ambiental Rural (CAR), a georeferenced registry covering 7 million rural properties and 700 million hectares, has been relaunched as Brazil's first official Digital Public Good during COP30, with new modules being developed to make the system more open, interoperable, and reusable⁸⁶. This combined with INPE's deforestation monitoring systems (DETER, PRODES) and satellite earth observation, gives Brazil a data stock that is genuinely difficult to replicate elsewhere⁸⁷.
- **Renewable energy advantage:** Over 80% of Brazil's electricity comes from renewable sources⁸⁸. This gives Brazil a comparative advantage for powering AI infrastructure with low carbon intensity directly relevant to data centre investment.
- **Critical minerals positioning:** Brazil holds significant reserves of materials critical to semiconductor manufacturing, battery storage and AI hardware including lithium, nickel, copper, graphite, rare earths, and niobium. The Ministry of Mines and Energy's 2026 investor guide explicitly links these reserves to higher value-added processing ambitions, framing them as strategic leverage rather than raw commodity exports⁸⁹.
- **EllaLink connectivity:** The 2021 EllaLink submarine fibre-optic cable connecting Brazil directly to Europe reduces dependence on North Atlantic data routes and lowers latency for transatlantic digital exchange representing a concrete step toward infrastructure diversification⁹⁰.

⁸² <https://www.gov.br/governodigital/pt-br/estrategias-e-governanca-digital/estrategianacional>

⁸³ <http://Gov.br>

⁸⁴ <https://www.bcb.gov.br/estabilidadefinanceira/pix>

⁸⁵ <https://www.gov.br/governodigital/pt-br/infraestrutura-nacional-de-dados/interoperabilidade/conecta-gov.br>

⁸⁶ <https://www.amazonfund.gov.br/en/projeto/Environmental-Regularization/>

⁸⁷ <https://infoamazonia.org/en/2022/02/15/prodes-and-deter-systems-against-deforestation-amazon/#:~:text=areas.satellite%20monitoring%20programs%20at%20INPE>

⁸⁸ <https://www.epe.gov.br/en/press-room/news/epe-publishes-the-summary-report-brazilian-energy-balance-2025?TermId=27727a54-baa8-4f4a-a9f6-761fa5d9dfe4&TermSetId=81034557-52ee-41fc-8006-a8f277b1655&TermStoreId=31423334-e3fc-4ff3-9956-0d09b48b681f&>

⁸⁹ <https://www.gov.br/mme/pt-br/assuntos/secretarias/geologia-mineracao-e-transformacao-mineral/guia-investidor-minerais-criticos/guia-investidor-estrangeiro-em-minerais-criticos/2026/investors-guide-en-us.pdf>

⁹⁰ <https://www.gov.br/mcom/pt-br/noticias/2021/junho/mais-conexao-cabo-submarino-que-conecta-o-brasil-a-europa-e-ativado>

- **Multilateral diplomatic positioning:** Brazil has leveraged the 2024 G20 presidency, 2025 BRICS presidency, and COP30 to project its climate-relevant systems outward through ideas such as Climate DPI and ClimateStack⁹¹. Additionally, it has signed a DPI MoU with India, participation in the 50-in-5 global campaign that supports 50 countries in building and governing core digital public infrastructure by 2028; partnered with Norway on CAR reform, and deepened digital cooperation with the EU through the EU-LAC Digital Alliance and the BELLA Programme^{92 93 94 95}.

Main Dependencies / Limits

- **Foreign cloud and advanced chips:** Brazil's compute base comprises 205 data centres, with AWS (São Paulo), Google Cloud, and Microsoft Azure operating locally, but it is not sovereign. Supercomputing capacity at LNCC's Santos Dumont and Petrobras' industrial HPC systems runs on imported NVIDIA GPUs and AMD/Intel CPUs. Brazil has no domestic advanced semiconductor production, and the ReData tax-incentive regime for data centres incentivised local investment but not local ownership of the hardware layer⁹⁶.
- **Fragmented implementation across a vast federal territory.** The brownfield strategy depends on interoperability between federal, state, and municipal systems, but implementation is uneven. Remote and rural areas also face weak connectivity and high infrastructure costs, with many areas still lacking broadband access, especially where private investment is less profitable, thus leaving such projects to be mostly dependent on public subsidies.
- **AI talent shortfall.** The country does have significant institutional capacity in areas such as digital government and environmental monitoring, with agencies such as INPE and LNCC playing important roles in science, computing and public data systems^{97 98}. But Brazil still faces shortages in advanced digital and AI skills. Brasscom estimated demand for 797,000 ICT professionals between 2021 and 2025, with a projected deficit of over 500,000 if training levels did not rise⁹⁹. Public-sector AI capability is still being built: MGI and national agencies launched a civil servant AI training track in 2024, especially in procurement, implementation, auditing, and long-term governance of complex digital systems.
- **Environmental tensions:** As digital infrastructure expands, its escalating energy and water demands are coming into sharper tension with the adverse effects

⁹¹ <https://itsrio.org/wp-content/uploads/2017/01/Digital-Public-Infrastructure-for-Climate-Report.pdf>

⁹²

<https://50in5.net/#:~:text=On%20November%208th%202023%2C%2050.journeys%20for%20digital%20public%20infrastructure.>

⁹³ <https://50in5.net/wp-content/uploads/2024/12/Brazil.pdf>

⁹⁴ https://www.eeas.europa.eu/eeas/eu-lac-digital-alliance-leading-human-centric-artificial-intelligence_en

⁹⁵ https://www.eeas.europa.eu/delegations/brazil/bella-programme_en

⁹⁶

<https://www.gov.br/mme/pt-br/a-revolucao-brasileira-em-energia-e-mineracao/energia-eletrica/redata-politica-de-acesso-e-incentivos-para-data-centers>

⁹⁷ <https://www.gov.br/inpe/pt-br>

⁹⁸ <https://www.gov.br/lncc/pt-br>

⁹⁹

<https://brasscom.org.br/estudo-da-brasscom-aponta-demanda-de-797-mil-profissionais-de-tecnologia-ate-2025/>

experienced by local traditional communities, making this contradiction increasingly difficult to govern.

- **Policy ambition without structural transformation:** Brazil's flagship policy instruments, the 2024 National Strategy for Digital Government, the PBIA 2024–2028 AI plan and the ReData tax-incentive regime are real commitments, but they stop short of addressing the deeper hardware dependency¹⁰⁰. PBIA proposes a new supercomputer and AI excellence centres, yet the effort remains oriented toward expanding access and national capability rather than becoming a domestic producer of cutting-edge chips or server infrastructure. ReData itself was enacted by Provisional Presidential Measure, expired in February 2026, and is currently awaiting Congressional approval.

Brazil's strategy is **brownfield DPI with diplomatic diversification**: build what is strategic domestically (DPI, climate data, energy infrastructure), reuse what already works rather than rebuilding from scratch, and diversify external ties to avoid lock-in by any single partner. The brownfield logic is efficient as it generates leverage from existing systems rather than requiring the massive capital outlays that greenfield approaches demand.

Why this strategy? Usage of Assets:

Brazil's strategy reflects the political and administrative realities of a large federal state with extensive legacy systems and uneven territorial capacity. Rather than pursuing a greenfield digital overhaul, Brazil prioritises upgrading and integrating existing systems into its **digital public infrastructure (DPI)**, allowing it to generate leverage from systems already embedded in public administration while avoiding the fiscal and political costs of rebuilding from scratch. Simultaneously, it leverages its **market power** and **physical endowments** to attract investment, while benefiting from its **diplomatic positioning** to diversify its partnerships and avoid dependencies.

¹⁰⁰<https://www.gov.br/mcti/pt-br/acompanhe-o-mcti/transformacaodigital/plano-brasileiro-de-inteligencia-artificial>

Comparison Matrix

The four cases reveal how middle powers prioritise different layers of the digital stack depending on market size, developmental needs, institutional capacity, and geopolitical positioning. While the underlying condition of structural dependence is shared, the strategies diverge in where leverage is built, which dependencies are accepted, and how actively states seek to renegotiate them. The matrix below summarises these configurations across a common set of variables.

Case	How it frames digital sovereignty	Main leverage	Main dependencies	Dominant strategy	Main limitation
EU	Rule-setting power and strategic autonomy: shaping the conditions under which digital technologies operate, while diversifying and building productive capacity to reduce dependencies	Regulatory market power (“Brussels Effect”); ASML semiconductor chokepoint; nuclear energy assets; standard-setting authority abroad through partnerships and exporting norms; large market and capital mobilisation; emerging diversification through Global Gateway and the International Digital Strategy	US cloud infrastructure (AWS, Azure, Google); no hyperscalers; talent drain; fragmented internal market; capital gap; weak frontier AI production; energy dependence; exposure to US strategic shifts	Regulatory-industrial balancing: use legal and normative power while trying to build compute, cloud, and AI capacity through massive public investment (AI Continent Action Plan, EuroStack); diversification of partnerships to reduce dependencies	Confused standard-setting with capability-building for too long; now risks dismantling regulatory leverage before industrial alternatives are ready
Kenya	Visibility, credibility, and governance influence built through operational DPI and attraction of foreign investment without ownership of the full stack	M-Pesa and DPI ecosystem; eCitizen; NOFBI; renewable electricity; PAPSS / PesaLink regional positioning; diplomatic visibility as East Africa’s digital hub	Foreign-owned compute and cloud (Microsoft/G42); imported chips; state data on US/Chinese clouds; weak data governance enforcement; no binding AI law; external capital; talent drain with no retention mechanisms	DPI-as-diplomatic-leverage: convert operational digital credibility into continental and global positioning, attracting investment while managing (not resolving) structural compute and data dependence	Hosting foreign infrastructure is not owning it; Kenya’s energy advantage attracts foreign capital but has not yet generated sovereign productive capacity

Case	How it frames digital sovereignty	Main leverage	Main dependencies	Dominant strategy	Main limitation
India	Techno-nationalist sovereign AI: framed as anti-colonial self-reliance ("Atmanirbhar Bharat"), build national DPI and AI capability to have control over key governance and application layers, while reducing dependence on US/Chinese platforms	India Stack DPI (Aadhaar, UPI, DigiLocker); Bhashini and multilingual data; IndiaAI Mission and public compute; global DPI, diplomacy; huge domestic market; talent depth; expanding strategic positioning for semiconductors	Semiconductors dependencies; foreign hyperscalers dominate cloud; talent drain; data protection weaker than localisation ambitions; reliance on foreign open-source foundation models	Greenfield DPI with selective sovereignty: build from scratch where strategically valuable, accept managed dependence on chips/cloud, and use DPI credibility to claim global governance leadership	The Atmanirbhar rhetoric of self-reliance systematically outpaces the structural reality of import dependence — sovereignty is claimed at the application layer while the hardware layer remains foreign
Brazil	Domestic DPI assets built reusing existing systems ("brownfield approach"), while diversifying external partnerships rather than accepting lock-in	Brownfield DPI at scale (Gov.br, Pix, Conecta Gov.br, CAR); EllaLink connectivity; renewable electricity; climate and environmental data; critical minerals reserves; multilateral diplomacy and diversified partnerships (G20, BRICS, COP30, EU-LAC Alliance, 50-in-5)	Foreign cloud and advanced chips (no domestic semiconductor production); uneven territorial implementation; weak connectivity in remote areas; AI talent deficit; politically contingent legislation; environmental tensions around infrastructure expansion	Brownfield DPI with diplomatic diversification: upgrade existing systems, link digital infrastructure to social, economic and climate missions, and use multilateral platforms to project model internationally while diversifying dependencies	Political contingency and implementation unevenness — the strategy is coherent but depends on administrative continuity in a volatile political environment and cannot yet close the compute sovereignty gap

A posture of smart interdependence

What emerges from this analysis is a family of approaches, each shaped by local capabilities and constraints, yet all recognisable as variations of the same underlying posture: **smart interdependence** (also known as **intelligent dependence**), understood as the deliberate identification, prioritisation, and governance of dependencies in ways that build leverage over time¹⁰¹. Not all dependencies carry the same strategic risk. Dependence on foreign semiconductors or cloud infrastructure carries fundamentally different strategic risks than dependence on foreign governance norms or application-layer platforms. The former is structurally difficult to exit, while the latter remains open to negotiation, regulation, or gradual substitution.

¹⁰¹

<https://www.govtechintelhub.org/case-study-details/better-stack-%28part-1%29:-10-theses-to-demystify-the-debate-on-digital-sovereignty-and-the-euro-stack/aJYTG0000000te14AA>

Keohane and Nye (1977) identified this logic decades ago, arguing that what matters in any relationship of interdependence is not the dependency itself, but who bears the higher cost of walking away from it¹⁰². The middle powers examined here are not trying to escape dependence. What the cases reveal, rather, is a pattern of prioritisation, with states investing in the layers where domestic capacity is feasible and leverage can be built, while accepting the structural dependencies they cannot yet address. Smart interdependence is therefore less a policy model than a strategic orientation, moving beyond both the illusion of full sovereignty and the passivity of unmanaged dependence, toward a posture in which states build leverage where they can, accept dependence where they must, and invest continuously in the capacity to renegotiate both. Recent debates frame this less as self-sufficiency than as a more collaborative form of sovereignty, achieved through structured cooperation rather than isolation¹⁰³.

Conclusion

The four cases examined here each represent a different configuration of smart interdependence. The EU wields regulatory market power to shape the rules governing technologies it does not build, buying time to develop productive capacity while risking the erosion of its normative leverage before industrial alternatives are ready. India builds public digital infrastructure from the ground up at the application layer while remaining structurally dependent on foreign chips, cloud, and foundation models at the hardware layer. Kenya converts the operational credibility of its public digital systems into diplomatic positioning and foreign investment, building the institutional profile of a digital leader while the underlying compute remains foreign-owned. Brazil upgrades existing systems, diversifying external partnerships and using critical minerals and renewable energy as strategic assets in what its own government terms as a brownfield approach, but also without controlling the hardware and cloud layers beneath it and with uneven state capacity across its territory.

None of these cases has achieved digital sovereignty in any absolute sense. What the cases collectively demonstrate is that digital sovereignty is not a condition to be reached but a capacity to be continuously built, defended, and renegotiated across five strategic levers (market power, regulatory authority, digital public infrastructure, physical endowments, and diplomatic positioning). Middle powers that invest strategically across these five dimensions are not sovereign in the traditional sense, but they are not passive either. Compute and semiconductor dependencies still stand out as the hardest to address and the most politically consequential, with none of the middle powers examined here meaningfully resolving them. However, they are accumulating the leverage needed to negotiate these dependencies rather than simply inheriting them.

¹⁰² [Keohane and Nye \(1977\)](#)

¹⁰³ <https://ecdpm.org/work/brazil-world-open-digital-infrastructure-climate-cooperation>